

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Quiz # 5

Date: November 28, 2025

Time: 10:30 AM - 10:50 AM

Total Points: 10

Duration: 20 minutes

Note:

- Please write your name and roll number in the space provided.
- There is only one problem carrying a total of 10 points. Do all your work ONLY in the space provided.
- This is an closed note and closed book examination.

Name: _____ Roll No: _____

1. For $n \geq 1$ let $X_n \sim \text{Poisson}(n)$. Show that as $n \rightarrow \infty$,

$$\frac{X_n - n}{\sqrt{n}} \xrightarrow{d} \text{Normal}(0, 1).$$